

AutoIntern

End-to-End Workflow Guide

AI-Powered Internship Platform · Student Registration to HR Delivery

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1 SYSTEM HIGH-LEVEL PIPELINE

Regardless of the application type, every user goes through a unified entry flow. The platform orchestrates AI parsing, continuous job scraping, and real-time matching to create a seamless experience from registration to job application.

1	Student Registers Uploads raw PDF/Word resume to the platform
2	AI Resume Parser FastAPI extracts text → Claude-3-Haiku converts to standardised JSON profile → saved to Supabase PostgreSQL
3	Background Scrapers Continuous scraping workers feed the jobs table with internship listings from company sites & RSS feeds
4	Real-Time Matching FastAPI compares student skills against all jobs using a fast, low-cost AI model to calculate a 0–100 Match Score
5	Student Job Feed Matched jobs appear in the personalised feed, ranked by relevance and Match Score

Common Setup Phases

- **Onboarding:** The student uploads a raw PDF/Word resume. The FastAPI backend extracts the text, sends it to *Claude-3-Haiku* to convert it into a standardised JSON profile, and writes it to the database.
- **Job Gathering:** A continuous scraping loop feeds the *jobs* table with internship listings from company sites and RSS feeds.
- **Matching:** When the student opens the job feed, FastAPI compares their skills with all available jobs in real-time, calculating a **0–100 Match Score** using a fast, low-cost AI model.

2 WORKFLOW TYPE 1: DIRECT EMAIL-BASED APPLICATIONS

This workflow is used when a job listing includes a direct email contact — common with small-to-mid-sized businesses and startups. The system handles AI tailoring, PDF generation, pixel-based tracking, and SMTP delivery end-to-end.

Technical Sequence — Step by Step

1	Job Selection The student ticks checkboxes next to up to 10 jobs on their personalised feed and clicks " Tailor Selected ".
2	AI Tailoring (Claude 3.5 Sonnet) For each job, the backend sends the student's base resume and job description to <i>Claude 3.5 Sonnet</i> . It rewrites bullet points to emphasise relevant skills (keeping underlying facts true) and drafts a tailored cold pitch.
3	Pixel Injection & PDF Generation The backend converts the rewritten resume JSON into a clean PDF. A unique <i>app_id</i> entry is created and an invisible 1×1 pixel tracking image is embedded in the email HTML. When the recruiter's client loads the email, this pixel fires a request to the tracking endpoint.
4	Draft Review The API returns all ready-to-review drafts to the React frontend. The student can inspect each tailored resume and cover letter before confirming.
5	Bulk Send The student clicks " Send All " and the backend dispatches emails through Resend or AWS SES using a professional SMTP router. Application status is updated to <i>applied</i> .
6	Recruiter Opens Email (Tracking) When the recruiter opens the email, their client fetches the invisible pixel from the FastAPI tracking endpoint <i>/track/open/{app_id}.png</i> . The endpoint registers the hit and updates the application status to <i>seen</i> .
7	Kanban Update The student's dashboard dynamically slides the application card into the " Seen " column in real time via a WebSocket or polling update.

Tracking Pixel — Email HTML Snippet

HTML

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The 1×1 pixel image is invisible to the recipient but is fetched by the email client upon opening. This allows the platform to detect when an HR recruiter has viewed the application.

3 WORKFLOW TYPE 2: PORTAL-BASED APPLICATIONS

This workflow is used when a company requires submissions through online platforms such as LinkedIn Jobs, Internshala, Workday, or Greenhouse. AutoIntern respects portal rules to prevent student accounts from being banned, so all AI tailoring happens before the student interacts with the portal.

Technical Sequence — Step by Step

1	Initiate Portal Apply The student clicks " Apply via Portal ". AutoIntern immediately triggers the AI tailoring pipeline in the background so assets are ready before the student touches the external portal.
2	AI Tailoring <i>Claude 3.5 Sonnet</i> generates a tailored resume PDF, a cover letter, and pre-written answers for standard portal essay fields (e.g., "Describe a complex React problem you solved").
3	Split-Screen Copilot Launch The frontend launches a dedicated split-screen viewer: the Left Side loads the external job portal (LinkedIn, Greenhouse, etc.) in an iframe or browser tab; the Right Side shows the AutoIntern Copilot panel with all pre-generated assets.
4	Download Tailored Resume The student clicks " Download Tailored Resume " on the right panel, then uploads the PDF to the portal's upload box on the left side.
5	Copy-Paste AI Answers The copilot displays pre-written responses for each portal essay field. The student clicks " Copy " and pastes directly into the portal text fields — reducing manual effort significantly.
6	Portal Submission The student clicks " Submit Application " on the external portal website to complete the submission.
7	Mark as Applied The student clicks " Mark as Applied " in the AutoIntern sidebar. The system registers the application with status <i>applied</i> and the Kanban board updates accordingly.

LEFT PANE External Portal (LinkedIn / Greenhouse / Workday)

RIGHT PANE AutoIntern Copilot (Tailored assets, Copy buttons)

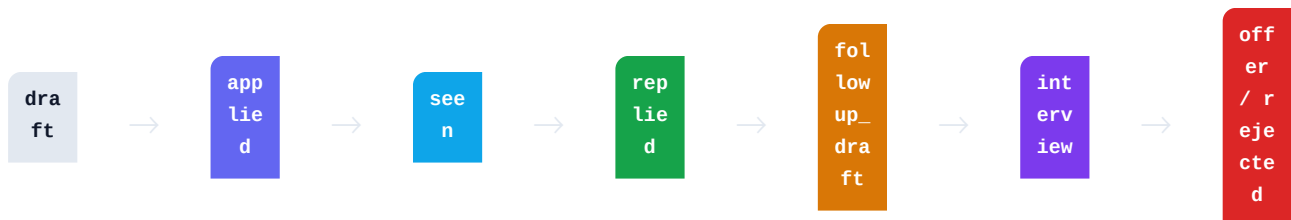
Split-screen Copilot interface layout

4 UNIFIED APPLICATION LIFECYCLE

Every application tracked in AutoIntern moves through a structured set of states. The lifecycle spans from initial AI tailoring through to final outcome logging, with automated triggers at each transition point.

Status	Trigger Event	System Action
<code>`draft`</code>	Student clicks "Tailor"	AI generates materials and caches them; user can review and edit before sending.
<code>`applied`</code>	Student clicks "Send" (Email) or "Mark as Applied" (Portal)	Moves card to "Applied" column; increments AI usage count for the student.
<code>`seen`</code>	HR Recruiter opens email (triggers tracking pixel)	Dynamically slides card to "Seen" column. Email workflow only — not available for portal.
<code>`replied`</code>	Recruiter replies to the outreach email	Auto-registered via incoming mail webhook; sends an alert notification to the student.
<code>`followup_draft`</code>	5 days pass with zero recruiter response	Background worker auto-drafts a follow-up email and places it in the student's queue.
<code>`interview`</code>	Student manually updates or AI scans a scheduling link	High-priority calendar alert is sent; the system provides tailored interview prep suggestions.
<code>`offer` / `rejected`</code>	Student manually updates the final outcome	System aggregates the data into the student's weekly analytics and performance feed.

Application State Flow



5 BACKEND DIRECTORY MAP (FASTAPI)

The following table maps each platform mechanic to its implementation file within the FastAPI backend. This serves as a quick reference for developers onboarding to the codebase.

Feature / Mechanic	File Path	Description
PDF Extraction & Formatting	<code>backend/app/services/ai_service.py</code>	Processes uploaded resumes — extracts text from PDF/Word and calls Claude-3-Haiku for JSON conversion.
Tracking Pixel Hook	<code>backend/app/api/applications.py</code>	Houses the <code>/track/open/{app_id}.png</code> route that detects email opens and updates application status.
Background Scraping Scheduler	<code>backend/app/workers/tasks.py</code>	Runs Celery / scheduler jobs to continuously fetch internship listings from company sites and RSS feeds.
Email SMTP Router	<code>backend/app/services/email_service.py</code>	Coordinates the Resend API connection for dispatching tailored email applications via professional SMTP.

Workflow Comparison Summary

	Email-Based Workflow	Portal-Based Workflow
Target Companies	SMBs, Startups	LinkedIn, Workday, Greenhouse, Internshala
AI Model	Claude 3.5 Sonnet	Claude 3.5 Sonnet
Delivery Method	Resend / AWS SES (SMTP)	Manual upload + copy-paste via Copilot
Open Tracking	✓ Pixel-based (real-time)	✗ Not available
Bulk Apply	✓ Up to 10 at once	✗ One at a time
Status Update	Automatic (pixel + webhook)	Manual ("Mark as Applied" button)

This document covers the complete AutoIntern workflow from student registration through to HR delivery. For implementation details, refer to the respective backend service files listed in Section 5. All AI operations use Claude 3.5 Sonnet for tailoring and Claude-3-Haiku for parsing.